

Prompt Stability in Arabic Toxicity Detection: A Comparative Evaluation of LLM Prompting Strategies

Ahmed Jlidi, Aya Khattab, Salma Soliman

Department of Computer Science

Kuwait University, Kuwait

s2231171736@ku.edu.kw, s2231172267@ku.edu.kw, s2211146689@ku.edu.kw

Abstract—Arabic toxicity detection is challenging because social media text often contains dialectal wording, informal spelling, code-mixing, sarcasm, political language, and culturally specific expressions. This paper studies prompt stability for Arabic toxicity classification using Gemini 3.0 Flash Preview on MAHED Task 1. We compare five prompt configurations and evaluate them using aggregate scores, per-class metrics, confusion matrices, prediction flip rate, McNemar’s test, and manual error analysis. The results show that prompt methods have similar overall scores but different behavior: broader prompts increase recall, while stricter prompts reduce hate over-detection at the cost of missing more true hate cases. Manual analysis links many prompt failures to label-boundary ambiguity, political criticism, code-mixing, implicit hate, and cultural or religious phrasing. These findings show that Arabic toxicity classification should not rely on a single prompt or one aggregate metric.

I. INTRODUCTION

Online toxicity detection is important for content moderation, but it remains difficult for Arabic. Arabic social media includes dialect variation, informal spelling, code-mixing, sarcasm, political language, and cultural expressions. Because of this, a text may contain strong or negative words without necessarily being hate speech, while some harmful meanings may be indirect.

Large language models can be used for toxicity detection without fine-tuning, which makes them practical when labeled data and computing resources are limited. However, recent work shows that LLM performance can change depending on the prompt. This means that prompt design should be evaluated directly, rather than treated as a fixed detail.

This work builds on prior multilingual hate speech prompting research [1], but shifts the focus from broad multilingual comparison to Arabic prompt stability. We evaluate how different prompt formulations affect Arabic harmful-content classification on MAHED Task 1 [2], where each text is classified as *hate*, *hope*, or *not_applicable*.

Our contributions are as follows:

- **Arabic-focused prompt stability evaluation.** We study how prompt wording changes model behavior in Arabic toxicity classification, instead of only asking which model performs best.
- **Deeper prompt comparison.** We compare prompt methods using aggregate scores, per-class results, confusion matrices, label flips, and statistical testing.
- **Arabic-specific error analysis.** We manually inspect cases involving label ambiguity, political criticism, dialectal wording, code-mixing, sarcasm, and cultural context to explain why prompts succeed or fail.

II. RELATED WORK

A. Datasets

Arabic harmful-content research still relies on a relatively small set of complementary resources. *So Hateful!* introduced a multi-label Arabic Twitter dataset annotated for offensive content, hate speech, emotion, and related attributes, while later work added a dedicated Arabic hate-target corpus and a bilingual Arabic–English dataset for emotions and hope speech [3]–[5]. MAHED consolidates these directions into a shared-task benchmark whose Task 1 directly matches our setting with the three labels *hate*, *hope*, and *not_applicable* [2]. For robustness analysis, functional benchmarks such as HateCheck and Multilingual HateCheck remain important because they test capabilities such as negation, quoted hate, counter-speech, and lexical confounds that are not captured by aggregate held-out metrics [6], [7]. Recent representative benchmarks such as HATEDAY further show that evaluations on curated academic datasets can substantially overestimate real-world moderation quality [8].

B. Multi-LLM Evaluations

Recent work has moved beyond single-model, single-prompt reporting. Roy et al. show that target information and explanatory context can substantially change zero-shot hate-speech detection performance [9]. Masud et al. show that persona and geographic framing alter

LLM moderation behavior across languages, which is directly relevant for culturally contextual prompting [10]. Zhang et al. identify systematic sensitivity biases in abusive-language detection, including under-prediction by instruction-tuned models and over-prediction by models shaped by human feedback [11]. Ghorbanpour et al. provide the closest setup to ours by comparing multiple prompting families across eight non-English languages and showing that no single prompt type dominates across languages and evaluation regimes [1]. More recent benchmark studies also suggest that reasonable label accuracy does not imply deeper alignment in explanation or moral reasoning [12].

C. Prior Methods

Two method families are most relevant. The first is supervised encoder fine-tuning, where multilingual encoders such as XLM-T and mDeBERTa remain strong baselines when sufficient labeled data is available [1]. The second is prompt-based inference, where definitions, few-shot examples, entailment formulations, role prompts, distinction prompts, and culturally informed instructions can each shift the effective decision rule at inference time [1], [9], [10]. Our work belongs to the second line, but narrows it to Arabic and to within-model prompt stability rather than broad cross-model ranking.

III. DATASET

This work uses the **MAHED 2025 shared task** dataset [5], [13], [14], a recent Arabic benchmark for harmful and prosocial content analysis. MAHED is built from real-world Arabic social media text and includes informal writing, dialectal variation, spelling variation, and mixed-script content.

In this work, we focus only on **Task 1**, which is a three-class text classification task. Each text is assigned one of three labels: *hate*, *hope*, or *not_applicable*. Other MAHED tasks are excluded because our goal is to evaluate prompt stability for text-only toxicity classification.

The dataset is suitable for this study for three reasons. First, it includes both harmful and constructive content, which allows us to test whether prompts can distinguish *hate*, *hope*, and neutral or irrelevant cases. Second, it is based on Arabic social media text, having 9,843 samples, which reflects noisy and informal Arabic usage. Third, the dataset was annotated by native Arabic speakers with high agreement of 0.85 using cohen kappa’s score, making it reliable for evaluation.

A. Task 1 Labels

Task 1 uses the following labels:

Hate: text that contains harmful, offensive, or discriminatory language directed toward a person or group.

Hope: text that expresses positive or constructive meaning, such as encouragement, optimism, support, or gratitude.

Table I: Example annotations from MAHED Task 1.

Text	Label
كل المهاجرين لصوص ويجب طردهم فوراً	hate
معاً يمكننا بناء مستقبل أفضل لأطفالنا	hope
اليوم هو يوم مشمس وجميل	not_applicable

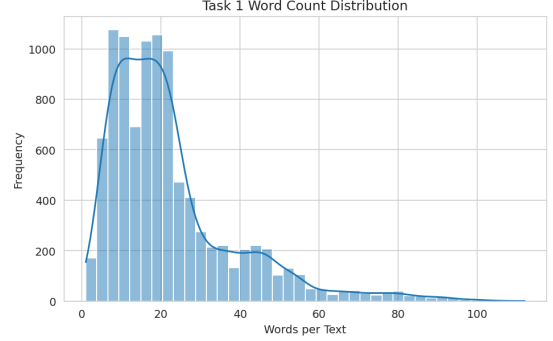


Figure 1: Distribution of text length in MAHED Task 1 measured by word count.

Not_applicable: text that does not clearly contain hate or hope. This includes neutral, irrelevant, unclear, or unrelated content.

Table I show several examples of MAHED task 1.

B. Descriptive Analysis

Figure 1 shows that most Task 1 texts are short, which is expected for social media data. Table II summarizes the main linguistic characteristics. The samples contain emojis, mentions, URLs, repeated punctuation, character elongation, and Latin-script text. This showcases the noisy nature of Arabic social media which makes the classification task more challenging.

IV. EXPERIMENTAL SETUP

In this section, we propose our research questions, hypotheses, we describe the model, sampling setup, prompts and evaluation metrics used in the experiment. The goal is to compare different prompting strategies under the same model and the same evaluation samples.

A. Research Questions and Hypotheses

This study evaluates how different prompting strategies affect Gemini 3 Flash on Arabic toxicity classification. The experiments use MAHED Task 1, which contains three labels: *hate*, *hope*, and *not_applicable*. The goal is to test whether prompt wording changes model performance and model behavior, especially for the *hate* class.

The research questions are:

- **RQ1.** How does prompt formulation affect Gemini 3 Flash performance on MAHED Task 1?
- **RQ2.** Do prompt methods differ in how strict they are when assigning *hate* and *hope* labels?

Table II: Linguistic characteristics of the MAHED Task 1 dataset.

Metric	Value
<i>Length</i>	
Average word count	22.38
<i>Social media markers</i>	
Texts with emoji (%)	14.02
Texts with hashtag (%)	4.91
Texts with mention (%)	24.92
Texts with URL (%)	8.49
<i>Informal writing patterns</i>	
Texts with repeated punctuation (%)	15.24
Texts with elongation (%)	13.92
Texts with Latin/code-switching (%)	37.36

Table III: Experimental Setup

Parameter	Value
Model	Gemini 3.0 Flash Preview
Datasets	MAHED Task 1
Subset size	500 samples
Sampling strategy	Stratified sampling
Batch size	25 samples per API request
Random seed	42

- **RQ3.** Does distinction-based prompting reduce hate over-detection compared with zero-shot definition prompting?
- **RQ4.** Do culturally guided and NLI-style prompts help with Arabic-specific ambiguity and semantic uncertainty?
- **RQ5.** What types of errors explain prompt failures and prompt disagreements?

The hypotheses are defined as follows. First, prompt formulation is expected to change both the overall performance and the per-class behavior of Gemini 3 Flash. Second, zero-shot definition prompting is expected to be broader and more recall-oriented, while distinction-based prompting is expected to be stricter and reduce hate false positives. Third, culturally guided prompting is expected to help with Arabic-specific cases involving dialect, cultural phrasing, sarcasm, or indirect meaning. Fourth, NLI-style prompting is expected to require stronger evidence before assigning a label. Finally, prompt errors are expected to be mainly caused by political criticism, code-mixing, implicit hate, and cultural or religious phrasing.

B. Model

We use Gemini 3.0 Flash Preview as the LLM baseline. This model was selected because it is available through the provided API, supports Arabic input, and can be used without local GPU resources. Since the goal of this study is to test prompt formulation, we keep the model fixed across all experiments.

The main experimental settings are summarized in Table III.

Temperature and top-p are set to 0 to ensure consistent, and reproducible outputs

C. Sampling Protocol

For each dataset, we create a fixed stratified subset of 500 samples. Stratified sampling is used to keep the label distribution close to the original dataset distribution. The selected subset is saved before inference and reused across all prompt settings. This makes the comparison fair because each prompt is tested on the same examples.

D. Prompt Configurations

We evaluate five prompt configurations. These prompts were selected because they represent the main prompt families used in prior hate-speech prompting work and because each one tests a different way of guiding the model decision.

The first prompt is **zero-shot definition prompting**. In this setting, the model receives the label names and short definitions, but no examples. This prompt is used as the main baseline because it gives the model the task labels in a simple and direct form.

The second prompt is **few-shot definition prompting**. This prompt uses the same label definitions as the zero-shot definition prompt, but it also includes labeled examples from the training set. This setting tests whether examples help the model better understand the boundary between *hate*, *hope*, and *not_applicable*.

The third prompt is **zero-shot NLI prompting**. In this setting, the task is framed as an entailment-style decision. Instead of only asking for the class, the prompt asks which label meaning is supported by the text. This setting tests whether semantic framing helps the model avoid relying only on surface words.

The fourth prompt is **zero-shot distinction prompting**. This prompt explains the difference between hate, general toxicity, offensive language, and unrelated or unclear text. This setting is included because Arabic political, emotional, or rude text may be confused with hate. The goal is to test whether clearer label boundaries reduce hate over-detection.

The fifth prompt is **three-shot cultural prompting**. This prompt includes labeled examples and adds Arabic-specific guidance about dialectal wording, informal expressions, sarcasm, political context, religious phrasing, and indirect meaning. This setting is included because Arabic toxicity is often context-dependent and may not be clear from literal wording alone.

All prompts use a strict output format. The model must return one line per input text using the sample index and the predicted label. We batch 25 samples in each request to reduce the number of API calls. This is needed because the free API access has a low request limit, around 20 requests per day. Batching reduces the number of requests while keeping the same number of evaluated samples.

Table IV: Overall prompt performance on MAHED Task 1.

Prompt	Accuracy	Macro Precision	Macro Recall	Macro-F1
Few-shot Definition	0.654	0.638	0.704	0.654
Three-shot Cultural	0.675	0.658	0.718	0.673
Zero-shot Definition	0.634	0.622	0.694	0.636
Zero-shot Distinction	0.682	0.693	0.607	0.631
Zero-shot NLI	0.654	0.642	0.708	0.656

E. Evaluation Metrics

We report both aggregate and per-class metrics. The aggregate metrics are accuracy, macro precision, macro recall, macro-F1, and invalid output rate. The per-class metrics are precision, recall, F1, and support for each class. We also report confusion matrices to show the main error patterns.

Macro-F1 is the main metric because accuracy alone can hide poor performance on minority toxic or hate classes. For prompt comparison, we report the change in macro-F1, the change in toxic or hate class recall, the change in toxic or hate class precision, and the change in invalid output rate.

F. Statistical Testing

Since all prompt settings are evaluated on the same subset, the comparison is paired. We use McNemar’s test to compare whether two prompt settings make different numbers of classification errors on the same samples. We also use bootstrap resampling to estimate confidence intervals for macro-F1. A result is treated as statistically significant when the p-value is below 0.05.

V. RESULTS

This section reports the MAHED Task 1 experiments. The goal is to evaluate whether prompt formulation changes Gemini 3.0 Flash Preview performance on Arabic toxicity classification. All prompt methods are evaluated on the same stratified subset of 500 validation samples. This makes the comparison paired, since each prompt is tested on the same text instances.

The evaluated prompt methods are zero-shot definition, few-shot definition, zero-shot NLI, zero-shot distinction, and three-shot cultural prompting. The labels are *hate*, *hope*, and *not_applicable*. We report aggregate metrics, per-class results, confusion matrices, prompt sensitivity measures, statistical comparison, and hate-class error patterns.

A. Aggregate Prompt Performance

Table IV reports the overall results for all prompt methods. Macro-F1 is used as the main metric because the dataset is imbalanced. Accuracy is also reported because it gives a general view of total correct predictions. Figure 2 visualizes the same aggregate metrics across the five prompt methods.

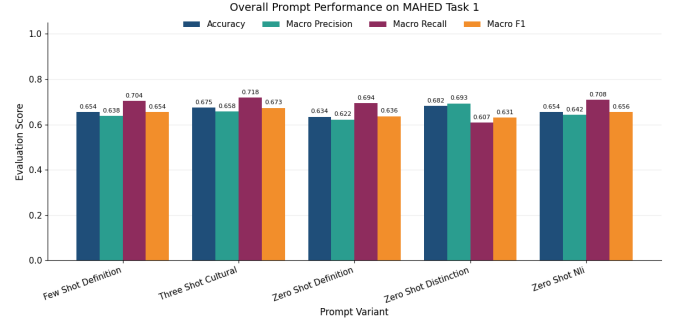


Figure 2: Overall prompt performance on MAHED Task 1 across the five prompting techniques.

B. Per-Class Performance

Figure 3 shows the precision, recall, and F1-score for each class under each prompt method. This allows comparison across the three labels: *hate*, *hope*, and *not_applicable*. Additionally,

Figure 4 shows the normalized confusion matrices for all prompt methods. The rows represent the gold labels, and the columns represent the predicted labels. These matrices show how often each gold class is assigned to each predicted class.

C. Statistical Prompt Comparison

We compare each prompt against zero-shot definition using McNemar’s test. This test is suitable because all prompts are evaluated on the same samples. The test compares the number of cases where zero-shot definition is correct and the other prompt is wrong against the number of cases where zero-shot definition is wrong and the other prompt is correct. Figure 5 reports the discordant counts and p-values for each comparison.

D. Prompt Sensitivity

We compute flip rate to measure prompt sensitivity. Flip rate is the percentage of samples where a prompt gives a different prediction from zero-shot definition. Figure 6 shows the flip rate for each prompt compared with zero-shot definition.

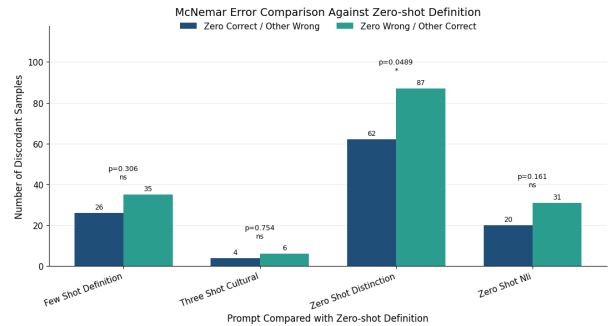


Figure 5: McNemar error comparison against zero-shot definition.

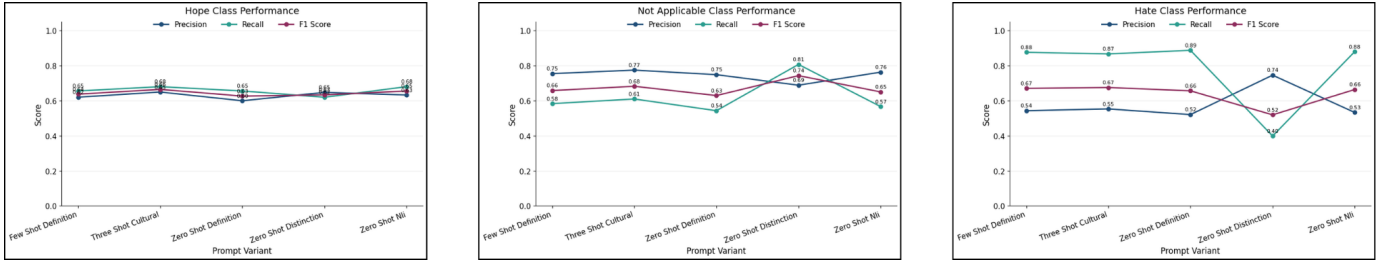


Figure 3: Per-class precision, recall, and F1 by prompt method.

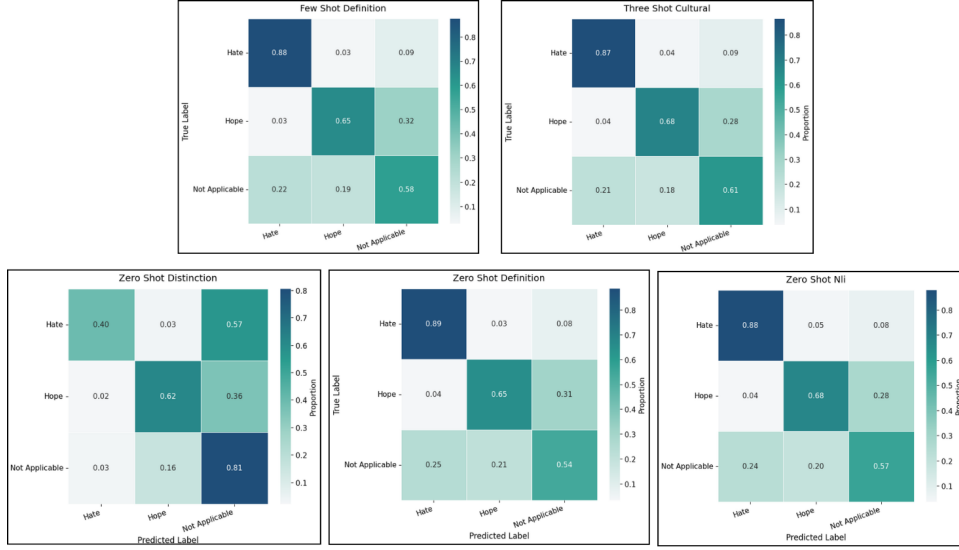


Figure 4: Normalized confusion matrices for all prompt methods.

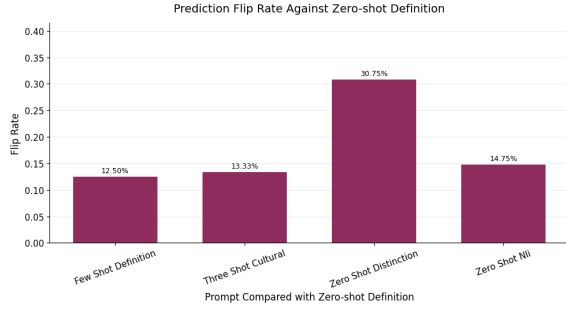


Figure 6: Prediction flip rate against zero-shot definition.

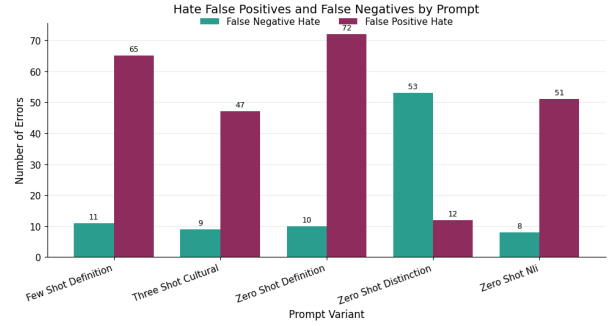


Figure 7: Hate false positives and false negatives by prompt method.

E. Hate-Class Error Analysis

Because *hate* is the most critical class in toxicity detection, we separately analyze hate false positives and hate false negatives. A hate false positive occurs when the model predicts *hate*, but the gold label is *hope* or *not_applicable*. A hate false negative occurs when the gold label is *hate*, but the model predicts another label. Figure 7 reports these two error types for each prompt.

VI. ANALYSIS AND DISCUSSION

A. Overall Prompt Behavior

Table IV and Figure 2 show that the prompt methods have close overall scores, but they do not behave in the same way. Three-shot cultural prompting obtains the highest macro-F1, while zero-shot distinction obtains the highest accuracy and macro precision. However, zero-shot distinction has lower macro recall. This suggests that aggregate metrics alone are not enough to explain prompt behavior.

B. Class-Level Differences

Figure 3 shows that the largest variation appears in the *hate* class. Some prompts achieve higher recall but lower precision, while others achieve higher precision but lower recall. This means that the prompts differ in how strict they are before assigning the *hate* label.

Zero-shot definition is more recall-oriented. As shown in Figure 3, it keeps hate recall high but has lower hate precision. This means it detects many true *hate* cases, but it also labels some non-hate cases as *hate*. In contrast, zero-shot distinction has higher hate precision but much lower hate recall, which means it is stricter before assigning the *hate* label.

C. Confusion Patterns

Figure 4 helps explain the per-class results. Broader prompts tend to classify more samples as *hate* or *hope*, which increases recall but also increases false positives. Stricter prompts reduce some false positives, especially for *hate*, but may miss true *hate* cases.

This shows that the best prompt depends on the target goal. If the goal is to catch more possible hate, a broader prompt may be useful. If the goal is to reduce over-detection, a stricter prompt may be better.

D. Statistical Comparison

Figure 5 shows the paired error comparison against zero-shot definition. Most prompt variants do not produce a statistically strong change compared with zero-shot definition. However, zero-shot distinction shows the strongest change. It fixes more zero-shot errors than it introduces new errors, and its result is close to the significance threshold.

This suggests that explicitly separating hate from general offensive or ambiguous content changes model behavior more than simply adding examples. This is also consistent with the flip-rate result in Figure 6, where zero-shot distinction changes the largest number of baseline predictions.

E. Prompt Sensitivity

Figure 6 shows that zero-shot distinction has the highest flip rate. This means it changes the largest number of predictions compared with zero-shot definition. Few-shot definition, three-shot cultural, and zero-shot NLI have lower flip rates, meaning they change the baseline predictions less often.

This supports the finding that distinction prompting has the strongest effect on the model decision boundary. In other words, the prompt that explicitly separates hate from related non-hate categories changes model behavior the most.

F. Hate-Class Error Trade-Off

Figure 7 shows a clear trade-off for the *hate* class. Zero-shot definition produces many hate false positives and few hate false negatives. This means it is aggressive in detecting hate. It catches many true hate cases, but it also over-labels non-hate text as hate.

Zero-shot distinction shows the opposite pattern in Figure 7. It produces very few hate false positives, but many hate false negatives. This means it is strict and avoids over-detection, but it misses more true hate cases. Zero-shot NLI and three-shot cultural prompting show a more balanced behavior, reducing some false positives while keeping false negatives relatively low.

G. Manual Error Analysis

To better explain the numerical results, we manually inspect two types of cases. First, we inspect shared failures, where all prompt methods fail on the same sample. These cases help identify hard examples that are difficult regardless of prompt design. Second, we inspect prompt-specific wins and losses, where one prompt is correct and another prompt is wrong. These cases help explain why prompts behave differently.

For the manual assessment, we use the following possible error causes:

- **Label-boundary ambiguity:** negative or emotional text that is not clearly hate or hope under the dataset label.
- **Dialectal wording:** dialect-specific words, spelling variants, or local expressions.
- **Sarcasm or irony:** the literal wording differs from the intended meaning.
- **Implicit hate or indirect attack:** harm is implied without explicit slurs.
- **Political criticism:** strong criticism of politicians, countries, parties, or movements that may be confused with hate.
- **Religious or cultural phrasing:** prayers, curses, religious references, or culturally loaded expressions.
- **Code-mixing or Arabizi:** Arabic mixed with English, French, or Romanized Arabic.
- **Annotation ambiguity:** the gold label itself is debatable or context-dependent.

Table V shows four representative shared failures. In these examples, all prompt methods produced the same wrong decision. This suggests that the errors are not caused by one prompt only, but by difficult cases such as ambiguous label boundaries, implicit meaning, political context, or cultural phrasing.

The shared-failure examples show that some errors are difficult for all prompt methods. Case 1 and Case 7 show that emotional or supportive wording can be confused with *hope*. Case 2 shows that implicit hate can be missed when the text does not contain a clear hateful expression.

Table V: Representative shared failures where all prompt methods failed.

Text	Manual assessment
مرأوا اول بطوله رسميه بلرن انتشافي! السور 6 مره في تاريخ بلرن! عودوا البلرن بعد غياب لا ثلاث سنين	Gold: not_applicable . All prompts predicted hope . Error cause: Label-boundary ambiguity . The text has positive sports emotion, so the prompts treated it as hope, although the dataset label is not_applicable.
لا أستطيع أن أنزل لدرجة هؤلاء الناس عندما أشعر أنهم ظلموني، أشعر بالغضب الشديد بسبب قسوة المشاعر والظلم	Gold: hate . All prompts predicted not_applicable . Error cause: Implicit hate or indirect attack . The harmful meaning is not explicit, so the prompts failed to assign the hate label.
♥ الله يوفقكم قلبنا معكم FBC	Gold: not_applicable . All prompts predicted hope . Error cause: Label-boundary ambiguity; Religious or cultural phrasing . The phrase contains support and religious wording, which made all prompts predict hope.
@aurasi @jebren صحراء سوف تحرروها فينكم قرنية هيا يا سيدكم اسرائيل ليبيا سادس	Gold: not_applicable . All prompts predicted hate . Error cause: Political criticism; Code-mixing or Arabizi . The political wording and mixed text caused the prompts to over-detect hate.

Case 10 shows that political and mixed-script text can be incorrectly pushed toward *hate*.

In addition to shared failures, we inspect prompt-specific wins and losses. These are cases where one prompt is correct while another prompt is wrong. Table VI summarizes the main cases where each prompt performs better than the others.

The manual analysis shows that prompt methods fail and succeed for different reasons. Zero-shot definition catches more possible *hate* and *hope*, but it may over-label ambiguous text. Few-shot definition helps with emotional or religious phrasing. Three-shot cultural prompting helps when Arabic-specific context is important. Zero-shot distinction is useful when the model must avoid confusing political criticism or rude language with hate. Zero-shot NLI helps when the label needs stronger semantic support.

H. Main Finding

Across Table IV and Figures 3–7, the main finding is that prompt methods differ mostly in strictness. Broad prompts increase recall but also increase false positives. Stricter prompts reduce over-detection but may miss true cases. Therefore, Arabic toxicity classification should not rely on only one prompt or one aggregate metric. Per-class behavior, prompt sensitivity, statistical testing, and error analysis are needed to understand how the model behaves.

I. Limitations

This study has three main limitations. First, the evaluation uses a stratified subset of 500 samples from MAHED Task 1. This size is enough for an initial prompt comparison, but a larger subset or the full validation/test split would give more reliable estimates.

Second, the experiments use one LLM. This keeps the analysis focused on prompt behavior, but the findings may not generalize to other models. Future work should compare multiple LLMs to check whether the same prompt patterns appear across different model families.

Third, the study tests several prompt methods separately, but it does not fully explore prompt combinations. The results suggest that different prompts have different strengths and weaknesses, and their performance depends on the context of the input.

VII. CONCLUSION

This paper evaluated prompt stability for Arabic toxicity classification using Gemini 3.0 Flash Preview on MAHED Task 1. We compared five prompt configurations under the same model and the same evaluation subset. The results show that prompt design affects not only overall performance, but also per-class behavior and error patterns.

The main finding is that prompts differ mostly in how strict they are before assigning *hate* or *hope*. Zero-shot definition is broader and catches more possible hate and hope cases, but it also creates more false positives. Zero-shot distinction is stricter and reduces hate over-detection, but it misses more true hate cases. NLI-style and cultural prompts show more balanced behavior in some cases, especially when meaning depends on semantic evidence or Arabic-specific context.

The manual error analysis shows that many errors are caused by label-boundary ambiguity, political criticism, implicit hate, code-mixing, dialectal wording, and religious or cultural phrasing. This supports the need to evaluate Arabic toxicity systems beyond a single score. Per-class metrics, confusion matrices, flip rate, statistical testing, and qualitative error analysis give a clearer view of model behavior.

Future work should evaluate larger subsets or the full MAHED split, compare multiple LLMs, and test hybrid prompts that combine clear label definitions, cultural guidance, and stricter rules for ambiguous or non-hateful negative text. Dialect-level analysis is also an important direction for improving Arabic toxicity detection.

Table VI: Summary of manual prompt-specific error analysis.

Winning Prompt	Wins	Labels in Wins	Dominant Case Characteristics
zero_shot_definition	40	hope: 15, NA: 13, hate: 12	label-boundary ambiguity; code-mixing; political criticism
few_shot_definition	10	NA: 7, hope: 3	label-boundary ambiguity; religious/cultural phrasing
three_shot_cultural	10	NA: 9, hope: 1	dialectal wording; political criticism; toxicity vs. hate
zero_shot_distinction	10	NA: 10	political criticism; toxicity vs. hate; religious/cultural phrasing
zero_shot_nli	10	NA: 6, hope: 3, hate: 1	toxicity vs. hate; code-mixing

REFERENCES

- [1] F. Ghorbanpour, D. Dementieva, and A. Fraser, “Can prompting LLMs unlock hate speech detection across languages? a zero-shot and few-shot study,” in *Proceedings of the 9th Workshop on Online Abuse and Harms (WOAH)*. Vienna, Austria: Association for Computational Linguistics, Aug. 2025, pp. 413–425. [Online]. Available: <https://aclanthology.org/2025.woah-1.39/>
- [2] W. Zaghouani, M. R. Biswas, M. Bessghaier, S. Ibrahim, G. Mikros, A. Hasnat, and F. Alam, “MAHED shared task: Multimodal detection of hope and hate emotions in Arabic content,” in *Proceedings of the Third Arabic Natural Language Processing Conference: Shared Tasks*. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 560–574. [Online]. Available: <https://aclanthology.org/2025.arabicnlp-sharedtasks.75/>
- [3] W. Zaghouani, H. Mubarak, and M. R. Biswas, “So hateful! building a multi-label hate speech annotated Arabic dataset,” in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. Torino, Italia: ELRA and ICCL, May 2024, pp. 15 044–15 055. [Online]. Available: <https://aclanthology.org/2024.lrec-main.1308/>
- [4] W. Zaghouani and M. R. Biswas, “An annotated corpus of Arabic tweets for hate speech analysis,” in *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era*. Varna, Bulgaria: INCOMA Ltd., Shoumen, Bulgaria, Sep. 2025, pp. 1413–1419. [Online]. Available: <https://aclanthology.org/2025.ranlp-1.163/>
- [5] —, “Emohopespeech: An annotated dataset of emotions and hope speech in english and arabic,” in *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing-Natural Language Processing in the Generative AI Era*, 2025, pp. 1406–1412.
- [6] P. Röttger, B. Vidgen, D. Nguyen, Z. Waseem, H. Margetts, and J. Pierrehumbert, “HateCheck: Functional tests for hate speech detection models,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Online: Association for Computational Linguistics, Aug. 2021, pp. 41–58. [Online]. Available: <https://aclanthology.org/2021.acl-long.4/>
- [7] P. Röttger, H. Seelawi, D. Nozza, Z. Talat, and B. Vidgen, “Multilingual HateCheck: Functional tests for multilingual hate speech detection models,” in *Proceedings of the Sixth Workshop on Online Abuse and Harms (WOAH)*. Seattle, Washington (Hybrid): Association for Computational Linguistics, Jul. 2022, pp. 154–169. [Online]. Available: <https://aclanthology.org/2022.woah-1.15/>
- [8] M. Tonneau, D. Liu, N. Malhotra, S. A. Hale, S. Fraiberger, V. Orozco-Olvera, and P. Röttger, “HateDay: Insights from a global hate speech dataset representative of a day on Twitter,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 2297–2321. [Online]. Available: <https://aclanthology.org/2025.acl-long.115/>
- [9] S. Roy, A. Harshvardhan, A. Mukherjee, and P. Saha, “Probing LLMs for hate speech detection: strengths and vulnerabilities,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 6116–6128. [Online]. Available: <https://aclanthology.org/2023.findings-emnlp.407/>
- [10] S. Masud, S. Singh, V. Hangya, A. Fraser, and T. Chakraborty, “Hate personified: Investigating the role of LLMs in content moderation,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 15 847–15 863. [Online]. Available: <https://aclanthology.org/2024.emnlp-main.886/>
- [11] Y. Zhang, V. Hangya, and A. Fraser, “LLM sensitivity challenges in abusive language detection: Instruction-tuned vs. human feedback,” in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 2765–2780. [Online]. Available: <https://aclanthology.org/2025.coling-main.188/>
- [12] J. Trager, F. Vargas, D. Alves, M. Guida, M. K. Ngueajio, A. Agrawal, Y. Daryani, F. K. Malekabadi, and F. M. Plaza-del Arco, “MFTCXplain: A multilingual benchmark dataset for evaluating the moral reasoning of LLMs through multi-hop hate speech explanation,” in *Findings of the Association for Computational Linguistics: EMNLP 2025*. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 15 709–15 740. [Online]. Available: <https://aclanthology.org/2025.findings-emnlp.851/>
- [13] W. Zaghouani, M. R. Biswas, M. Bessghaier, S. Ibrahim, G. Mikros, A. Hasnat, and F. Alam, “Mahed shared task: Multimodal detection of hope and hate emotions in arabic content,” in *Proceedings of the third arabic natural language processing conference: shared tasks*, 2025, pp. 560–574.
- [14] W. Zaghouani and M. R. Biswas, “An annotated corpus of arabic tweets for hate speech analysis,” in *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing-Natural Language Processing in the Generative AI Era*, 2025, pp. 1413–1419.